Identifying morality features in language models

Allen G hart

March 28, 2026

1 Introduction

Possible framing:

- Mechanistic interpretability aims to identify human-understandable variables inside language models.
- Prior work has found linear directions for sentiment and other properties, but morality-like representations are less settled, especially in small open models.
- The core question here is not whether Qwen behaves morally in some philosophical sense, but whether a simple, stable, causally relevant linear direction separates ETHICS scenarios labeled acceptable vs unacceptable/wrong.
- The paper's main claim can be modest: in Qwen2.5-0.5B-Instruct, a layer-15 residual direction fit from ETHICS short commonsense scenarios is stable, predictive, causally relevant, and only partly explained by sentiment.

Draft contribution list.

Something like:

- We identify a simple layerwise mean-difference direction in raw residual activations that separates ETHICS commonsense scenarios.
- We show that this direction is strongest around layer 15 and stable under both increasing sample size and bootstrap resampling.
- We show causal relevance by ablating the direction and measuring a substantial AUROC drop on moral judgment prompts.
- We compare this direction to a negative-sentiment direction learned from SST-2 and show moderate alignment but clear angular separation.
- We show that a sentiment-controlled residual morality direction still has causal force, and that random-direction ablations do not produce comparable effects.

2 Related Work

This section can stay short and tightly scoped. Likely subsections:

- Mechanistic interpretability and linear directions.
- Sentiment directions and representation engineering.
- Morality / ethics benchmarks in language models.
- Prior "moral direction" work and why the present paper is narrower and cleaner.

Use the literature note already in `notes/morality_direction_literature_note.md`.

Citation cheat sheet.

These are placeholder citation notes so you can see the available references and decide where each belongs while writing.

[**Zou et al.(2023)Zou, Phan, Chen, Campbell, Guo, Ren, Pan, Yin, Mazeika, Dombrowski, Goel, Li, Byun,**
Representation engineering establishes the general methodology of extracting high-level linear directions from hidden states and using them for analysis and intervention.

[**Tigges et al.(2024)Tigges, Hollinsworth, Geiger, and Nanda**]
Sentiment is represented approximately linearly in activation space, with causal intervention validating the direction.

[**Hendrycks et al.(2021)Hendrycks, Burns, Basart, Critch, Li, Song, and Steinhardt**]
ETHICS introduces a large benchmark of shared human value judgments, including the commonsense morality subset used here.

[**Schramowski et al.(2022)Schramowski, Turan, Andersen, Rothkopf, and Kersting**]
An early "moral direction" paper showing that pre-trained language models contain a moral or normative direction in embedding space.

[**Hämmerl et al.(2023)Hämmerl, Deiseroth, Schramowski, Libovický, Rothkopf, Fraser, and Kersting**]
Multilingual language models show language-dependent moral bias, which is relevant for any future cross-lingual extension.

[**Zhou et al.(2024)Zhou, Yu, Zhang, Xu, Huang, and Li**]
Safety alignment and jailbreak behavior can be studied through intermediate hidden states, providing adjacent precedent for hidden-state analysis of ethical behavior.

[**Liu et al.(2024)Liu, Mao, Tang, and Johnson**]
Self-correction can improve moral behavior at the output level without necessarily removing problematic internal representations.

[**Liu et al.(2025)Liu, Qi, Zhang, Cheng, and Johnson**]
Moral self-correction is not an innate capability, reinforcing the distinction between latent representations and surface behavior.

[**Hu et al.(2026)Hu, Zeng, Yang, and Lin**]
The closest recent prior work explicitly probes internal moral paths and steerable moral vectors in LLMs.

[**Xu et al.(2026)Xu, Rusnak, and Kaplan**]
Ethical framework representations appear structured and entangled, raising the possibility that morality is multi-dimensional rather than a single axis.

[**Socher et al.(2013)Socher, Perelygin, Wu, Chuang, Manning, Ng, and Potts**]
The original sentiment treebank paper introducing the SST dataset family used here as the sentiment control source.

[**Clifford et al.(2015)Clifford, Iyengar, Cabeza, and Sinnott-Armstrong**]
The original Moral Foundations Vignettes paper provides short, foundation-specific moral scenarios plus non-moral social-norm controls.

[**Crone et al.(2021)Crone, Rhee, and Laham**]
The brief MFV paper is useful background for the six-foundation extension and confirms a 90-item six-foundation variant exists.

[**Abdulhai et al.(2024)Abdulhai, Serapio-García, Crepy, Valter, Canny, and Jaques**]
A behavioral MFT study showing that LLMs express moral-foundation biases and that prompting can shift those expressed values.

[**Yu et al.(2026)Yu, Yi, Karimi-Malekabadi, Abdurahman, Ye, Narayanan, Zhao, and Dehghani**]
The closest foundation-level mechanistic prior art, studying how moral foundations are encoded in hidden states and partially disentangled across layers.

2.1 Mechanistic Interpretability and Linear Directions

Mention work on sparse autoencoders, representation engineering, sentiment directions, and the broader idea that useful internal variables may be approximately linear in hidden-state space.

2.2 Moral and Ethical Evaluation in Language Models

Explain that ETHICS and related datasets are mostly behavioral benchmarks. The novelty here is not a new moral dataset, but using ETHICS as a source of high-consensus scenarios for hidden-state analysis.

2.3 Closest Prior Work

Be explicit and honest:

- broad claim "models have internal moral representations" is not new;
- the contribution here is a residual-stream, small-open-model, layerwise, causal, sentiment-controlled study using ETHICS commonsense short scenarios.

3 Experimental Setup

3.1 Model

Current model:

- `Qwen/Qwen2.5-0.5B-Instruct`
- 24 layers
- layer 15 selected by held-out ETHICS performance from a layer sweep

You can motivate the model as a small, cheap, open model suitable for rapid mechanistic experiments, while acknowledging its language/comprehension limits.

3.2 Moral Dataset

ETHICS commonsense short scenarios.

- Label 1: unacceptable / wrong / immoral in the dataset framing.
- Label 0: acceptable / not wrong.
- Training counts used in the main run: 2789 immoral, 3872 acceptable.
- Held-out eval: 983 immoral, 1126 acceptable.

Be careful with terminology. In prose, you can note that the dataset is about consensus moral acceptability judgments rather than deep moral philosophy.

3.3 Sentiment Control Dataset

SST-2, with the sign flipped so the learned control direction points toward negative sentiment. Matched-size train subset:

- 2789 negative
- 2789 positive

This gives a fairer comparison than letting sentiment use a much larger dataset.

3.4 Activation Readout

Use the final token of a fixed prompt template, e.g.

- ETHICS representation prompt: `Scenario: {text}\nMoral evaluation:`
- sentiment representation prompt: `Text: {text}\nSentiment:`

The rationale is that the final prompt token is a natural readout position where the preceding scenario has been integrated.

4 Methods

4.1 Direction Estimation

The simplest possible estimator:

$$d = \mu_{\text{immoral}} - \mu_{\text{acceptable}}$$

with d normalized to unit length. Stress the simplicity: no SAE, no nonlinear probe, no multi-token aggregation beyond the final pooled state.

4.2 Layerwise Sweep

Describe the layer sweep over raw residual activations. The main result:

- best ETHICS eval AUROC around layer 15
- nearby mid/late layers also strong

Possible figure: layerwise AUROC plot.

4.3 Held-Out Evaluation

The basic question is whether a direction fit on the train split separates acceptable vs immoral scenarios on held-out ETHICS examples. Likely metrics:

- AUROC as the primary metric
- accuracy as secondary, with a note that it is threshold-sensitive

4.4 Causal Intervention

Define the intervention clearly. If u is the learned unit direction and h the hidden state at the prompt boundary token, the removal intervention is:

$$h' = h - (h \cdot u)u$$

The judgment prompt scores `acceptable` vs `unacceptable`. Explain why AUROC is the right primary outcome: it measures separation, not threshold calibration.

4.5 Sentiment Control

Three comparisons belong here:

- full immorality direction
- pure negative-sentiment direction
- immorality residual after orthogonalizing against sentiment

Write down the orthogonalization:

$$u_\perp = \frac{u - (u \cdot s)s}{\|u - (u \cdot s)s\|}$$

where u is immorality and s is the sentiment direction.

4.6 Random-Direction Null

Important control: matched-norm random-direction ablations at the same layer and token position. This addresses the objection that "any disturbance" might hurt performance.

4.7 Bootstrap Direction Uncertainty

This section can be short but useful. Explain the stratified bootstrap over training activations:

- resample each class with replacement at original class size
- refit the mean-difference direction
- compare the bootstrap direction to the full-data direction via cosine and angle

This yields interpretable angular stability estimates for both immorality and negative sentiment, plus the uncertainty of their alignment.

5 Results

5.1 A Stable Linear Immorality Direction Exists

Core numbers to include:

- full ETHICS sweep best layer: 15
- eval AUROC around 0.829 for the direction-only classifier
- sample-size stabilization already visible as train size increases

Good figure candidates:

- layerwise AUROC
- sample-size stability curve

5.2 Removing the Direction Degrades Moral Separation

Use the full held-out causal numbers. Main AUROC result:

- baseline: 0.7758
- remove immorality direction: 0.6357

State plainly that this is evidence of causal relevance, not merely correlation.

5.3 Sentiment Explains Part, But Not All, of the Signal

Key numbers:

- remove pure sentiment: AUROC 0.6607
- remove sentiment-orthogonalized immorality residual: AUROC 0.7407
- cosine between full-data immorality and negative-sentiment directions: 0.5219
- angle between them: about 58.5 degrees

The wording here matters. They are substantially aligned relative to random high-dimensional directions, but robustly distinct.

5.4 Random Directions Do Not Reproduce the Effect

Main null result:

- morality removal AUROC drop: about 0.140
- 32 random-direction removals: mean drop near 0, max drop about 0.007

This is one of the strongest controls in the paper.

5.5 Bootstrap Stability

Key numbers:

- immorality direction stays within roughly 6 degrees of the full-data estimate under bootstrap resampling
- negative-sentiment direction also within roughly 6 degrees
- the alignment between them is itself stable

Use the note in `notes/bootstrap_uncertainty.md`.

5.6 Visualization and Error Analysis

Possible figure set:

- immorality vs residual PC1 scatter
- same plot with negative-sentiment comparison arrow

Possible analysis paragraph:

- highest-projection immoral examples are extreme violence / abuse / fraud
- high-projection acceptable examples often contain violent or taboo lexical content
- low-projection immoral examples are often milder, oddly phrased, or linguistically noisy

This suggests the direction captures real moral signal, but is strongly shaped by vivid harm-related language.

5.7 Latent-Output Mismatch

Include the result that the latent immorality projection aligns with ETHICS labels slightly better than Qwen's explicit one-word judgment. This is a nice payoff because it suggests that the latent direction can be cleaner than the model's expressed answer in a weak model.

5.8 Foundation Directions Are Layered and Overlapping

This subsection can present the MFRC extension using the revised taxonomy rather than Haidt's original six:

- Care peaks early at layer 4.
- Loyalty and proportionality peak at layer 6.
- Authority peaks later at layer 19.
- Equality and purity peak later still at layer 22.

The core story is not "six orthogonal moral axes." It is that foundation-relevant directions are recoverable, peak at different depths, and remain partially overlapping when compared within a shared focal layer.

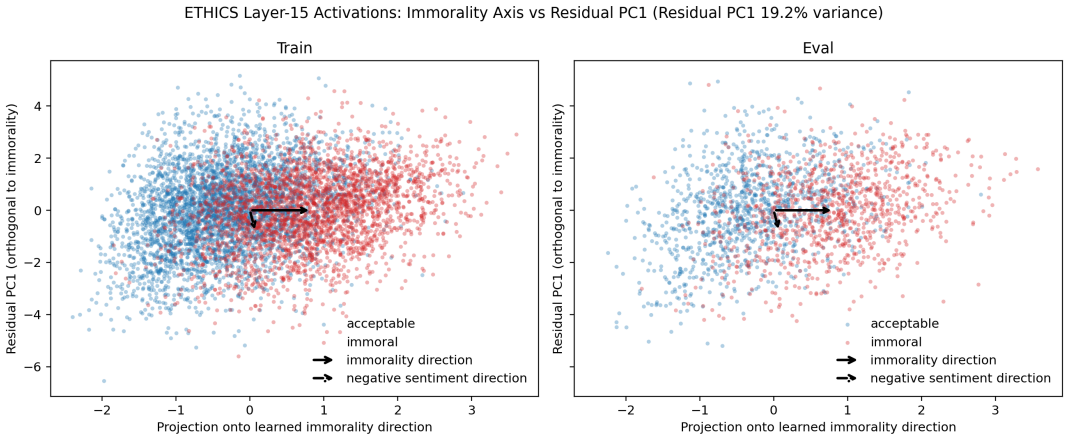


Figure 1: Layer-15 ETHICS activations projected onto the learned immorality direction and the leading residual principal component orthogonal to it. Blue and red points denote acceptable and immoral ETHICS scenarios, respectively. The solid arrow shows the learned immorality direction, while the dashed arrow shows the negative-sentiment direction from SST-2 projected into the same plane.

Row	Care	Loyalty	Proportionality	Authority	Equality	Purity
Care (L4)	–	29.4 [26.1, 40.8]	22.8 [20.8, 35.5]	52.5 [46.8, 61.9]	33.1 [29.4, 43.7]	23.6 [22.8, 38.6]
Loyalty (L6)	32.8 [28.6, 46.4]	–	31.2 [28.8, 45.2]	40.9 [34.4, 55.7]	32.9 [30.3, 46.9]	35.0 [31.8, 48.8]
Proportionality (L6)	26.1 [22.6, 39.7]	31.2 [28.8, 45.2]	–	48.4 [41.4, 61.2]	25.7 [24.0, 39.5]	36.2 [31.8, 50.1]
Authority (L19)	60.3 [50.0, 71.5]	38.9 [30.1, 54.3]	45.1 [35.3, 58.0]	–	29.7 [25.6, 41.2]	56.1 [45.7, 68.4]
Equality (L22)	48.8 [41.1, 59.1]	34.6 [32.7, 43.6]	34.5 [29.0, 46.5]	41.4 [34.2, 52.6]	–	45.6 [38.5, 56.8]
Purity (L22)	22.4 [21.0, 29.5]	39.6 [32.8, 52.6]	31.3 [28.3, 41.5]	71.1 [60.8, 82.2]	45.6 [38.5, 56.8]	–

Table 1: Angles between MFRC foundation directions evaluated in each focal foundation's best-decoding layer. Each row uses the focal foundation's best layer, and each cell reports the angle to another foundation direction fitted in that same layer, with bootstrap 95% confidence intervals. Smaller angles indicate stronger alignment.

6 Discussion

6.1 What Does the Direction Capture?

Good discussion point: the direction is neither a full moral theory nor just sentiment. It seems to combine:

- harm / violence
- taboo / crime semantics
- negative affect
- some additional moral signal beyond pure sentiment

6.2 Limitations

Likely limitations:

- small model with clear comprehension limitations
- one dataset and one prompt format
- linear direction is only an approximation
- ETHICS captures consensus labels, not universal morality
- residual-stream analysis alone does not identify a full circuit

Row	Care	Loyalty	Proportionality	Authority	Equality	Purity
Care (L4)	0.871	0.767	0.765	0.751	0.843	0.792
Loyalty (L6)	0.857	0.783	0.779	0.773	0.866	0.769
Proportionality (L6)	0.857	0.783	0.779	0.773	0.866	0.769
Authority (L19)	0.799	0.680	0.695	0.777	0.805	0.730
Equality (L22)	0.822	0.770	0.717	0.765	0.886	0.800
Purity (L22)	0.822	0.770	0.717	0.765	0.886	0.800

Table 2: Held-out AUROC for each MFRC foundation direction evaluated in the same focal best-decoding layers as Table 1. This provides the decoding context for the angular comparisons: some foundations remain strongly decodable even in layers where another foundation is maximal.

6.3 Future Work

Natural next steps:

- repeat on Gemma
- extend to controversial items and latent/output comparisons
- connect the direction to SAE features or circuits
- compare pretrained vs instruct-tuned models

7 Conclusion

Keep the conclusion modest and crisp. Something like:

- there is a simple, stable, causally relevant linear direction in Qwen 0.5B that tracks ETHICS moral wrongness / unacceptability judgments
- this direction is related to but not reducible to negative sentiment
- small models already contain morally relevant latent variables that can be studied mechanistically

A Prompt Templates

Put all representation and judgment templates here.

B Additional Tables

Useful appendix tables:

- layerwise metrics
- causal AUROC comparison table
- bootstrap summary table

C Example Error Cases

Pull examples from `notes/error_modes.md` and `notes/latent_expressed_mismatch.md`.

References

[Abdulhai et al.(2024)Abdulhai, Serapio-García, Crepy, Valter, Canny, and Jaques] Marwa Abdulhai, Gregory Serapio-García, Clement Crepy, Daria Valter, John Canny, and Natasha Jaques. Moral foundations of large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17737–17752, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.982. URL https://aclanthology.org/2024.emnlp-main.982/.

[Clifford et al.(2015)Clifford, Iyengar, Cabeza, and Sinnott-Armstrong] Scott Clifford, Vijeth Iyengar, Roberto Cabeza, and Walter Sinnott-Armstrong. Moral foundations vignettes: A standardized stimulus database of scenarios based on moral foundations theory. *Behavior Research Methods*, 47(4):1178–1198, 2015. doi: 10.3758/s13428-014-0551-2. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC4780680/.

[Crone et al.(2021)Crone, Rhee, and Laham] Damien L. Crone, Joshua J. Rhee, and Simon M. Laham. Developing brief versions of the moral foundations vignettes using a genetic algorithm-based approach. *Behavior Research Methods*, 53(3):1179–1187, 2021. doi: 10.3758/s13428-020-01489-y. URL https://pubmed.ncbi.nlm.nih.gov/33006066/.

[Hämmerl et al.(2023)Hämmerl, Deiseroth, Schramowski, Libovický, Rothkopf, Fraser, and Kersting] Katharina Hämmerl, Bjoern Deiseroth, Patrick Schramowski, Jindřich Libovický, Constantin Rothkopf, Alexander Fraser, and Kristian Kersting. Speaking multiple languages affects the moral bias of language models. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 2137–2156, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.134. URL https://aclanthology.org/2023.findings-acl.134/.

[Hendrycks et al.(2021)Hendrycks, Burns, Basart, Critch, Li, Song, and Steinhardt] Dan Hendrycks, Collin Burns, Steven Basart, Andrew Critch, Jerry Li, Dawn Song, and Jacob Steinhardt. Aligning AI with shared human values. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=dNy_RKzJacY.

[Hu et al.(2026)Hu, Zeng, Yang, and Lin] Luoming Hu, Jingjie Zeng, Liang Yang, and Hongfei Lin. The straight and narrow: Do LLMs possess an internal moral path?, 2026. URL https://arxiv.org/abs/2601.10307.

[Liu et al.(2024)Liu, Mao, Tang, and Johnson] Guangliang Liu, Haitao Mao, Jiliang Tang, and Kristen Johnson. Intrinsic self-correction for enhanced morality: An analysis of internal mechanisms and the superficial hypothesis. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 16439–16455, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.918. URL https://aclanthology.org/2024.emnlp-main.918/.

[Liu et al.(2025)Liu, Qi, Zhang, Cheng, and Johnson] Guangliang Liu, Zimo Qi, Xitong Zhang, Lu Cheng, and Kristen Johnson. Moral self-correction is not an innate capability in language models. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 660–683, Mumbai, India, December 2025. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-ijcnlp.39. URL https://aclanthology.org/2025.findings-ijcnlp.39/. Published version of the arXiv preprint "Self-correction is Not An Innate Capability in Language Models" (arXiv:2410.20513).

[Schramowski et al.(2022)Schramowski, Turan, Andersen, Rothkopf, and Kersting] Patrick Schramowski, Cigdem Turan, Nico Andersen, Constantin A. Rothkopf, and Kristian Kersting. Large pre-trained language models contain human-like biases of what is right and wrong to do. *Nature Machine Intelligence*, 4(3):258–268, 2022. doi: 10.1038/s42256-022-00458-8. URL https://www.nature.com/articles/s42256-022-00458-8.

[Socher et al.(2013)Socher, Perelygin, Wu, Chuang, Manning, Ng, and Potts] Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pages 1631–1642, Seattle, Washington, USA, October 2013. Association for Computational Linguistics. URL https://aclanthology.org/D13-1170/.

[Tigges et al.(2024)Tigges, Hollinsworth, Geiger, and Nanda] Curt Tigges, Oskar J. Hollinsworth, Atticus Geiger, and Neel Nanda. Language models linearly represent sentiment. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 58–87, Miami, Florida, US, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.blackboxnlp-1.5. URL https://aclanthology.org/2024.blackboxnlp-1.5/. Published version of the arXiv preprint "Linear Representations of Sentiment in Large Language Models" (arXiv:2310.15154).

[Xu et al.(2026)Xu, Rusnak, and Kaplan] Weilun Xu, Alexander Rusnak, and Frederic Kaplan. Probing ethical framework representations in large language models: Structure, entanglement, and methodological challenges, 2026. URL https://arxiv.org/abs/2603.23659.

[Yu et al.(2026)Yu, Yi, Karimi-Malekabadi, Abdurahman, Ye, Narayanan, Zhao, and Dehghani] Chenxiao Yu, Bowen Yi, Farzan Karimi-Malekabadi, Suhaib Abdurahman, Jinyi Ye, Shrikanth Narayanan, Yue Zhao, and Morteza Dehghani. Tracing moral foundations in large language models, 2026. URL https://arxiv.org/abs/2601.05437.

[Zhou et al.(2024)Zhou, Yu, Zhang, Xu, Huang, and Li] Zhenhong Zhou, Haiyang Yu, Xinghua Zhang, Rongwu Xu, Fei Huang, and Yongbin Li. How alignment and jailbreak work: Explain LLM safety through intermediate hidden states. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 2461–2488, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.139. URL https://aclanthology.org/2024.findings-emnlp.139/.

[Zou et al.(2023)Zou, Phan, Chen, Campbell, Guo, Ren, Pan, Yin, Mazeika, Dombrowski, Goel, Li, Byun, Wa] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency, 2023. URL https://arxiv.org/abs/2310.01405.